Reyhaneh Hosseininejad

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PhD candidate in Robotics at EPFL working on 3D human motion: pose estimation, forecasting, and physics-based generation and evaluation of human-human interaction, with physics simulation (Newton / NVIDIA Warp) and reinforcement learning.

Research Experience

Visual Intelligence for Transportation Lab (VITA) — 2022 – *present*
Doctoral Assistant | *EPFL, Switzerland*

Physics-Based Human Motion

- Simulating human-human interactions in **Newton (NVIDIA Warp)** for physics-informed motion generation, and training humanoid control policies with **reinforcement learning** (ProtoMotions with Newton; PHC with **Isaac Gym**).
- Developing metrics that use **physics simulation** to judge whether generated human-human motion is plausible (paper in preparation).
- Co-first author of **HHI-Assist** (RA-L 2025, presented at ICRA), a motion-capture dataset and benchmark of close-contact assistance between people, with a Transformer-based diffusion baseline. Now extending it to robot-assistance experiments in simulation.

Motion Forecasting and 3D Vision

- Co-first author of **MotionMap** (CVPR 2025), a heatmap representation that captures several plausible futures and their uncertainty, reaching state-of-the-art results in human motion forecasting.
- Used cross-task consistency between direct 3D estimation and 2D-to-3D lifting to improve 3D human pose estimation.
- Used TrajNet++ to forecast the movement of professional hockey players from positional data.

Other Projects and Mentorship

- Built a pipeline to create a large Persian speech dataset and fine-tuned an **LLM-based text-to-speech** model on it with multi-GPU training, using both full fine-tuning and LoRA.
- Early experiments toward a **world model**: connecting a Newton-based driving simulator to a video generation model.
- Working with the NeuroRestore lab on a virtual-patient model that maps spinal cord stimulation to muscle (EMG) responses, for spinal cord injury research, using MyoSuite on **MuJoCo**.
- Supervised Master's semester projects on SMPL-based 3D human modeling, 3D car pose estimation, and lane detection.

Center for Advanced Systems and Technologies (CAST) — 2017 – 2022
Research Assistant | *University of Tehran*

- Trained a **reinforcement learning** walking policy for the **Surena IV** humanoid in PyBullet with a reward that includes a **motion-imitation** term, including custom Gym environments, trajectory generation, and inverse kinematics (MS thesis) [Code].
- Simulated walking patterns for Surena IV in the Choreonoid simulator (BS research).
- Designed and built a motorized **prosthetic hand**, including its embedded electronics and EMG-based control (BS final project).

Selected Publications

MotionMap: Representing Multimodality in Human Pose Forecasting — CVPR 2025
Reyhaneh Hosseininejad *, M. Shukla *, S. Saadatnejad, M. Salzmann, A. Alahi

HHI-Assist: A Dataset and Benchmark of Human-Human Interaction in Physical Assistance — RA-L 2025
S. Saadatnejad *, *Reyhaneh Hosseininejad* *, J. Barreiros, K. Tsui, A. Alahi *(presented at ICRA)*

Physics-Derived Plausibility Metrics for 3D Human-Human Motion Generation — In Preparation
Reyhaneh Hosseininejad, T. Van Der Meer, M. Shukla, A. Alahi

PoseDriver: A Unified Approach to Multi-Category Skeleton Detection for Autonomous Driving Assistance — Under Review
Y. Borhani, T. Mordan, Y. Wang, *Reyhaneh Hosseininejad*, J. Khoramdel, A. Alahi

* *Equal contribution*

Additional Research Experience

Applied Artificial Intelligence Lab — 2020 – 2022
Research Assistant | *University of Tehran*

- Started a project on detecting keratoconus from corneal topography images.

Research Center for Biomedical Technologies and Robotics (RCBTR) — 2016
Research Intern | *IK Hospital Complex, Tehran*

- Designed calibration phantoms for micro CT-scans.
- Designed 3D-printed molds and made silicone breast-tumor models for biopsy training; developed training tools for the Sina surgical robot.

Education

PhD **EPFL**, EDRS (Robotics, Control, and Intelligent Systems) — 2022 – *present*
Thesis: 3D Human Motion: From Multimodal Forecasting to Physics-Informed Generation
Advisor: Prof. Alexandre Alahi
Expected graduation: 2027 (flexible)

MS **University of Tehran**, Mechanical Engineering, CGPA: 19.02 / 20.0 (4.0 / 4.0) — 2019 – 2022
Thesis: Controlling a Biped Robot with Reinforcement Learning
Advisors: Prof. Aghil Yousefi-Koma and Prof. Massoud Shariatpanahi

BS **University of Tehran**, Mechanical Engineering, CGPA: 16.89 / 20.0 (3.6 / 4.0) — 2014 – 2019
+ Minor in Computer Engineering

Teaching

Teaching Assistant — *EPFL*
Deep Learning for Autonomous Vehicles; Introduction to Machine Learning for Engineers — 2022 – 2026

Teaching Assistant — *University of Tehran*
Artificial Intelligence; System Identification; Measurement Systems and Instrumentation; Introduction to Mechanical Engineering — 2017 – 2021

Skills and Additional Information

Robot Learning & Simulation: Reinforcement Learning, Motion Imitation, ProtoMotions, PHC, Isaac Gym, Newton, NVIDIA Warp, MuJoCo, MyoSuite, PyBullet, Gym, Choreonoid
Programming: Python, C++, C
Machine Learning & Vision: PyTorch, TensorFlow, Hugging Face, Diffusion Models, Transformers, LLM Fine-Tuning (full-parameter, LoRA), Multi-GPU Training, SMPL/SMPL-X
Tools: Linux, Git, Docker, MATLAB, SolidWorks
Languages: English (Fluent), Persian (Native), French (A2)
Service: Reviewer, IEEE/CVF International Conference on Computer Vision (ICCV), 2025